A NEW ANGLE ON SAS

MARCUS LARSEN

1. Introduction

Neutral geometry is built on an axiomatic system that does not assume a parallel postulate. The axioms of Neutral Geometry, listed below, were chosen with such regard. Of these six axioms, the Side-Angle Side Congruency Condition (SAS) (Axiom 6) is particularly helpful in establishing the concept of triangle congruency in Neutral Geometry. It is seen as the foundational congruency condition whereby all remaining congruency conditions (including ASA, SSS, and AAS) are proven.

This paper investigates the role of SAS as an axiom within Neutral Geometry, as well as whether or not Neutral Geometry necessitates SAS to be the "foundational congruency condition."

2. SAS as an Axiom

2.1. **Understanding Axioms.** An axiomatic system is a framework in which all conclusions are deduced from a set of foundational assumptions, known as axioms. Axioms are the basis of an axiomatic system. All assumptions relevant to understanding a geometric model must be consistent with the axioms. Axioms are, by definition, assumed as true in their axiomatic system without needing to be proven; if they could be proven, they would be theorems.

There are six widely-accepted axioms of Neutral Geometry. Axiom 6 is most pertinent to this investigation. These six axioms provide the framework for Neutral Geometry, ensuring that all models of Neutral Geometry are consistent and allow for logical deduction.

Axiom 1 (The Existence Postulate)**.** *The collection of all points forms a nonempty set. There is more than one point in that set.*

The Existence Postulate ensures that Neutral Geometry begins with the fundamental assumption that points exist. It provides an understanding for the undefined concept of "points," which is foundational for constructing and comparing geometric objects.

Axiom 2 (The Incidence Postulate)**.** *Every line is a set of points. For every pair of distinct points A and B there is exactly one line l such that $A \in l$ and $B \in l$.*

The Incidence Postulate introduces the undefined concept of "lines" and their relationship to points, another crucial concept for geometric objects. These lines will be seen as the sides of the triangles discussed later.

Theorem 1. *If l and m are two distance, nonparallel lines, then there exists exactly one point P such that P lies on both l and m.*

In axiomatic systems, theorems like Theorem 1 are proven by nothing except the axioms themselves. Once proven, theorems can be used like axioms to prove future theorems. This particular Theorem will be seen in the later proofs for ASA from SAS and SAS from ASA.

Axiom 3 (The Ruler Postulate)**.** *For every pair of points P and Q there exists a real number PQ, called the distance from P to Q. For each line l there is a one-to-one correspondence from l to $\mathbb{R}$ such that if P and Q are points on the line that correspond to the real numbers x and y, respectively, then $PQ = \|x - y\|$.*

The Ruler Postulate established the concept of measurable distance. Without a way to measure and compare the distances of line segments, side congruence could not be established.

Axiom 4 (The Plane Separation Postulate)**.** *For every line l, the points that do not lie on l form disjoint, nonempty sets H_1 and H_2, called half-planes bounded by l, such that the following conditions are satisfied:*

(1) *Each of H_1 and H_2 is convex.*
(2) *If $P \in H_1$ and $Q \in H_2$, then $\overline{PQ}$ intersects l.*

The Plane Separation Postulate provides the framework for understanding geometric relationships, allowing for the separation of space into distinct regions. This separation is crucial in defining and comparing angles, which will be further defined in the following axiom.

Axiom 5 (The Protractor Postulate)**.** *For every angle $\angle BAC$ there is a real number $\mu(\angle BAC)$, called the measure of $\angle BAC$, such that the following conditions are satisfied:*

(1) $0^\circ \leq \mu(\angle BAC) < 180^\circ$ *for every angle* $\angle BAC$.
(2) $\mu(\angle BAC) = 0^\circ$ *if and only if* $\overrightarrow{AB} = \overrightarrow{AC}$.
(3) *For each real number r, $0 < r < 180$, and for each half-plane H bounded by $\overleftrightarrow{AB}$ there exists a unique ray $\overrightarrow{AE}$ such that E is in H and $\mu(\angle BAE) = r^\circ$*
(4) *If the ray $\overrightarrow{AD}$ is between rays $\overrightarrow{AB}$ and $\overrightarrow{AC}$, then $\mu(\angle BAD) + \mu(\angle DAC) = \mu(\angle BAC)$*

The Protractor Postulate is for angle measures what the Ruler Postulate is for distance measures. This postulate allows for the precise measurement and comparison of angles, a requirement for demonstrating angle congruence.

Axiom 6 (Side-Angle-Side Congruence Condition (SAS))**.** *If $\triangle ABC$ and $\triangle DEF$ are two triangles such that $\overline{AB} \cong \overline{DE}$, $\angle ABC \cong \angle DEF$, and $\overline{BC} \cong \overline{EF}$, then $\triangle ABC \cong \triangle DEF$.*

SAS is the focal point of this paper. SAS establishing the concept of triangle congruency in Neutral Geometry. It is seen as the foundational congruency condition whereby all remaining congruency conditions (including ASA, SSS, and AAS) are proven. In theory, if ASA, SSS, or AAS were able to both independently establish triangle congruency and prove SAS, then SAS would not need to be an axiom of Neutral Geometry. But this will be discussed at greater length later. First, it is important to understand the historical development of SAS in order to determine why SAS is given primacy above the other congruency conditions.

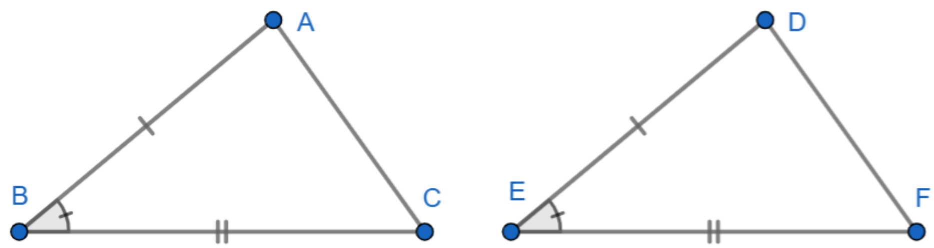

FIGURE 1. A model of SAS congruency

2.2. **History of SAS.** Euclid, in his 4th axiom, stated:

Things that coincide with one another are equal to one another.

This means that if there existed two triangles, $\triangle ABC$ and $\triangle DEF$, both triangles could be superimposed onto each other to prove that they were equal. Euclid noted that when the superimposed triangles were equal to one another (congruent), their corresponding sides would be equal, their corresponding angles would be equal, and their areas would be equal. This led Euclid to write Proposition 4:

> **Proposition IV:** *If two triangles have two sides equal to two sides respectively, and if the angles contained by those sides are also equal, then the remaining side will equal the remaining side, the triangles themselves will be equal areas, and the remaining angles will be equal, namely those that are opposite the equal sides.*

This states that, for $\triangle ABC$ and $\triangle DEF$, as long as any two corresponding sides and the angle between them were congruent, the triangles were congruent. This is SAS in its earliest form.

Contemporary mathematicians who have followed Euclid's example and written their own axiomatic systems have also weighed in on triangle congruency. David Hilbert (1862-1943) interestingly wrote his axiomatic system without reference to distance or angle measurements. Nonetheless, his fifth axiom of congruence states:

> **Axiom III-5:** *If for two triangles $\triangle ABC$ and $\triangle A'B'C'$ the congruences $\overline{AB} \cong \overline{A'B'}$, $\overline{AC} \cong \overline{A'C'}$, and $\angle BAC \cong \angle B'A'C'$ hold, then the congruence $\angle ABC \cong \angle A'B'C'$ is also satisfied.*

Hilbert was able to define the product of SAS – that is, congruency of another angle based on the conditions of SAS – without stating the SAS axiom. Hilbert's version essentially made fewer claims about the triangle's congruency but retained that SAS does have value in stating congruency.

George Birkhoff (1884-1944) notably did define congruency in his short, four-axiom axiomatic system. He replaced congruency with his famous Postulate of Similarity which states:

> **Postulate of Similarity:** *If in two triangles $\triangle ABC$, $\triangle A'B'C'$, and for some constant $k > 0$, $d(A', B') = kd(A, B)$, $d(A', C') = kd(A, C)$ and also $\angle B'A'C' = \pm\angle BAC$, then also $d(B', C') = kd(B, C)$, $\angle C'B'A' = \pm\angle CBA$, and $\angle A'C'B' = \pm\angle ACB$.*

In this way, Birkhoff defines the concept of similarity in his axiomatic method, emphasizing the proportionality between corresponding sides of two triangles to the scaling constant $k > 0$. Notice that in cases where $k = 1$, similar triangles have equal corresponding sides and are therefore congruent. In these cases, the distance of two sides and the measure of the inside angle is used to prove congruency between the two triangles. This is SAS. Saunders MacLane (1909-2005) used the same approach in his axiomatic system, too.

The School Mathematics Study Group (SMSG) developed its enormous 22-axiom-long axiomatic system in the 1960's. SMSG does make a direct statement about congruency, saying:

> **Postulate 15 (The S.A.S. Postulate):** *Given a correspondence between two triangles (or between a triangle and itself). If two sides and the included angle of the first triangle are congruent to the the corresponding parts of the second triangle, then the correspondence is a congruence.*

The University of Chicago School Mathematics Project (UCSMP) is not as direct as its successor, SMSG, in defining congruency. USCMP notably uses the Reflection Postulate to bypass the necessity of any congruence axiom.

2.3. **Why SAS is an Axiom.** The above sources, from Euclid to SMSG, use SAS as their foundational congruency axiom. However, the commonality of SAS as an axiom should not lead people the believe that SAS is inherently superior to the other congruency conditions. Below are some possible hypotheses as to why SAS became the dominant congruency axiom:

(1) **Logical Independence.** SAS is an axiom in neutral geometry, meaning it does not rely on any other geometric assumptions or postulates – it is logically independent. However, if another congruency condition, like ASA, was proven to be a suitable axiom, then SAS would no longer necessarily hold such primacy over the other congruency conditions. The question of SAS's unique logical independence will be explored in the final section of this paper.

(2) **Simplicity.** SAS is simply a simple congruency condition. Its three components are all adjacent to one another and it only requires knowledge of one angle measure. However, it should be noted that SAS these factors do not necessarily make SAS uniquely simple. Other congruency conditions, such as ASA, also has three adjacent components. Although ASA does require knowledge of two angle measures instead of one, it only requires knowledge of one side measure instead of two. Thus, ASA could technically be equally as "simple" as SAS in terms of amount of information needed.

(3) **Historical Precedent.** Probably the most prominent reason that SAS is used as the foundational congruence axiom is that it has held this title for centuries, beginning with Euclid's work. Its long-standing use as an axiom in classical geometry has made it a widely accepted and familiar starting point for geometric reasoning. The historical precedent contributes to its status as an axiom, as it has been well-tested in the development of classical geometry. However, it is important to remember that this historical preference doesn't mean that SAS is necessarily the "best" congruency condition. Other conditions, such as ASA, could have been adopted as the foundation in a logically consistent system.

(4) **Real-life Application.** SAS is often the easiest to use in real-life geometric construction, as it is usually easier to measure a second side than a second angle. This is a realistic grievance against ASA: it is too difficult to measure its second angle compared to SAS's second side. This practicality has led to SAS being frequently chosen as an axiom, as it is highly applicable in real-life constructions. However, as was said about it's historical precedent, real-life application does not necessarily make SAS "better" or even more logical than other congruency conditions.

3. Congruency Proofs from SAS

There are, as has been stated, other congruency conditions for triangles. These congruency conditions can each be proven using SAS as an axiom.

3.1. **ASA from SAS.** The ASA Theorem is another congruency condition. It is essentially the "inverse" of SAS where, instead of using the congruency of two corresponding sides and their included angle to define congruency, ASA uses the congruency of two corresponding angles and their included side.

Theorem 2 (Angle-Side-Angle Congruence Condition (ASA))**.** *If $\triangle ABC$ and $\triangle DEF$ are two triangles such that $\angle CAB \cong \angle FDE$, $\overline{AB} \cong \overline{DE}$, and $\angle ABC \cong \angle DEF$, then $\triangle ABC \cong \triangle DEF$.*

Proof. Let $\triangle ABC$ and $\triangle DEF$ be two triangles such that $\angle CAB \cong \angle FDE$, $\overline{AB} \cong \overline{DE}$, and $\angle ABC \cong \angle DEF$ (hypothesis). We must show that $\triangle ABC \cong \triangle DEF$.

There exists a point C' on $\overrightarrow{AC}$ such that $\overline{AC'} \cong \overline{DF}$ (Point Construction Postulate). Now $\triangle ABC' \cong \triangle DEF$ (SAS) and so $\angle ABC' \cong \angle DEF$ (definition of congruent triangles). Since $\angle ABC \cong \angle DEF$ (hypothesis), we can conclude that $\angle ABC \cong \angle ABC'$. Hence $\overrightarrow{BC} = \overrightarrow{BC'}$ (Protractor Postulate, Part 3). But $\overrightarrow{BC}$ can only intersect $\overrightarrow{AC}$ in at most one point (Theorem 1), so $C = C'$ and the proof is complete. □

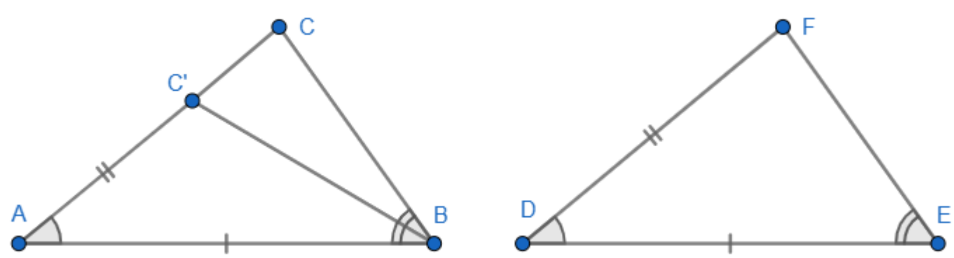

Figure 2. One possible location for C' in proof of ASA

3.2. **AAS from SAS.** Besides ASA, there are two other popular congruency conditions: AAS and SSS. Both of these theorems can be logically verified by SAS (Axiom 6) and ASA (Axiom 2).

Theorem 3. *If $\triangle ABC$ and $\triangle DEF$ are two triangles such that $\angle ABC \cong \angle DEF$, $\angle BCA \cong \angle EFD$, and $\overline{AC} \cong \overline{DF}$, then $\triangle ABC \cong \triangle DEF$.*

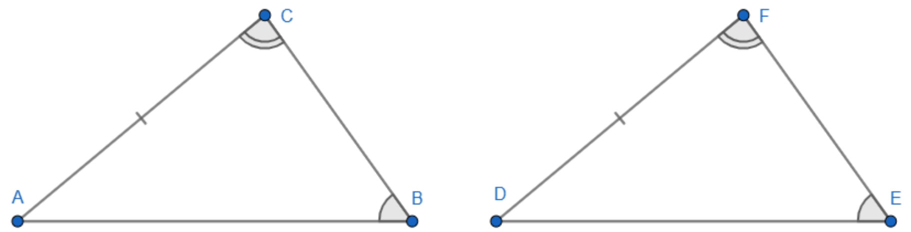

FIGURE 3. A model of AAS congruency

The following sample proof for AAS is one demonstrates this. Note that the proof below uses the Angle-Sum Theorem which, for the sake of brevity, has not been proven in this paper.

It is also noteworthy that Angle-Sum is unique to Euclidean Geometry and is not compatible with Neutral Geometry as whole. Thus, this is not a universally-applicable proof; note, however, that proofs for AAS from SAS do exist without reliance on Angle-Sum. This does make for an opportune spot to mention that the congruency conditions are not themselves subject to a particular parallel postulate, but are rather universal to Neutral Geometry.

Proof. Let $\triangle ABC$ and $\triangle DEF$ be two triangles such that $\angle ABC \cong \angle DEF$, $\angle BCA \cong \angle EFD$, and $\overline{AC} \cong \overline{DF}$ (hypothesis). We must show that $\triangle ABC \cong \triangle DEF$.

By the Angle-Sum Theorem, the third angles $\angle CAB$ and $\angle FDE$ are also congruent. Therefore, we now have two angles and the included side congruent for both triangles:

$$\angle CAB \cong \angle FDE,\ \overline{AC} \cong \overline{DF}, \text{ and } \angle ABC \cong \angle DEF.$$

This satisfies the ASA congruence condition, which has been proven above (Theorem 2). Thus, $\triangle ABC \cong \triangle DEF$. $\square$

3.3. SSS from SAS.

Theorem 4. *If $\triangle ABC$ and $\triangle DEF$ are two triangles such that $\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, and $\overline{CA} \cong \overline{FD}$, then $\triangle ABC \cong \triangle DEF$.*

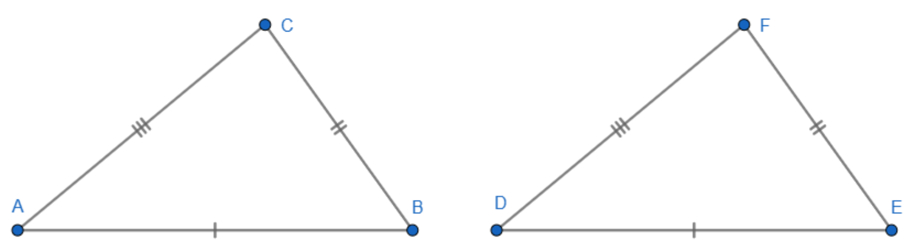

FIGURE 4. A model of SSS congruency

The following is a sample proof for SSS. Like the proof for ASA (Theorem 3), SSS may be proven using SAS (Axiom 6), ASA (Theorem 2), and AAS (Theorem

3), since all of these have already been logically verified. This proof also references another theorem (the Transitive Property of Congruence) which, for the sake of brevity, is not included in this paper.

Proof. Let $\triangle ABC$ and $\triangle DEF$ be two triangles such that $\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, and $\overline{CA} \cong \overline{FD}$ (hypothesis). We must show that $\triangle ABC \cong \triangle DEF$.

Construct $\triangle A'B'C'$ such that $\overline{A'B'} \cong \overline{DE}$, $\overline{B'C'} \cong \overline{EF}$, and $\angle A'B'C' \cong \angle DEF$. By SAS, $\triangle A'B'C' \cong \triangle DEF$. Since $\overline{A'C'} \cong \overline{FD}$ by construction and $\triangle A'B'C'$ is congruent to $\triangle ABC$, it follows that $\triangle ABC \cong \triangle DEF$ (Transitive Property of Congruence). $\square$

4. Another Approach

As long as another congruency condition can be assumed and used to independently prove SAS, then SAS could be replaced as axiom. As stated above, ASA is essentially the "inverse" of SAS. Both congruency conditions are based on having congruency of two corresponding parts and an included part. This makes it an ideal candidate to replace SAS as an axiom. So, for the sake of demonstration, ASA will temporarily replace SAS as Axiom 6, and SAS will temporarily replace ASA as Theorem 2. Thus:

Axiom 6 (Angle-Side-Angle Congruence Condition (ASA))**.** *If $\triangle ABC$ and $\triangle DEF$ are two triangles such that $\angle CAB \cong \angle FDE$, $\overline{AB} \cong \overline{DE}$, and $\angle ABC \cong \angle DEF$, then $\triangle ABC \cong \triangle DEF$.*

Theorem 2 (Side-Angle-Side Congruence Condition (SAS))**.** *If $\triangle ABC$ and $\triangle DEF$ are two triangles such that $\overline{AB} \cong \overline{DE}$, $\angle ABC \cong \angle DEF$, and $\overline{BC} \cong \overline{EF}$, then $\triangle ABC \cong \triangle DEF$.*

As seen above, ASA has been "promoted" to the level of an axiom (specifically as Axiom 6, as if it has replaced SAS). Likewise, SAS has been "demoted" to the level of a theorem (Theorem 2, as if it has replaced ASA). For this to logically work, the new ASA axiom will have to be used to prove the new SAS theorem. Remember that axioms are the foundation of an axiomatic systems, and that theorems can only be verified by previously-proven theorems. The following proof will demonstrate this:

Proof. Let $\triangle ABC$ and $\triangle DEF$ be two triangles such that $\overline{AB} \cong \overline{DE}$, $\angle ABC \cong \angle DEF$, and $\overline{BC} \cong \overline{EF}$ (hypothesis). We must show that $\triangle ABC \cong \triangle DEF$.

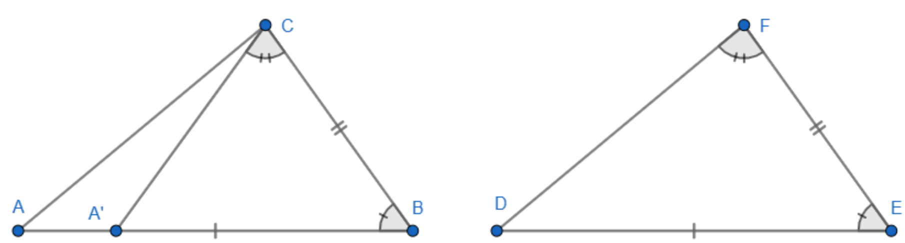

Figure 5. One possible location for A' in proof of SAS

There exists a point A' on $\overrightarrow{AB}$ such that $\angle BCA' \cong \angle EFD$ (Point Construction Postulate). Now $\triangle A'BC \cong \triangle DEF$ (ASA) and so $\overline{A'B} \cong \overline{DE}$ (definition of congruent triangles). Since $\overline{AB} \cong \overline{DE}$ (hypothesis), we can conclude that $\overline{AB} \cong \overline{A'B}$. Hence $\overrightarrow{CA} = \overrightarrow{CA'}$ (Protractor Postulate, Part 3). But $\overrightarrow{CA}$ can only intersect $\overrightarrow{AB}$ in at most one point (Theorem 1), so $A = A'$ and the proof is complete. □

It must be noted that the above proof for SAS from ASA is essentially the "inverse" of the proof for ASA from SAS. Rather than constructing a point such that the new side is congruent to the corresponding *side* of the second triangle, this proof constructs a point such that the new angle is congruent to the corresponding *angle* of the second triangle. However, this proof is still logical. Thus, it can be concluded that ASA can effectively prove SAS and therefore SAS holds no inherent primacy over the other congruency conditions.

It should be noted here that the remaining congruency conditions (AAS and SSS) do not need to be proven from the new ASA axiom. Remember that previously-proven theorems may be used to prove subsequent theorems. Since SAS is now a valid theorem, it can therefore be used to prove the validity of the other two as well. Both of these proofs will look identical to the ones seen earlier.

5. Conclusion

Throughout history, SAS has held a special place as the "foundational congruency condition" of Neutral Geometry. However, a thorough investigation of its axiomatic status has revealed that SAS holds few, if any, unique logical traits that give it primacy over other congruency conditions. It is not necessarily more simple than other conditions, requiring arguably the same amount of information as ASA. Likewise, it is no more logically independent than ASA which, as demonstrated in this paper, can be used as an axiom to prove an SAS theorem. As a result, while SAS may hold a historical and conventional significance, its status as the "foundational congruency condition" of Neutral Geometry appears logically arbitrary.